Lecture 21: Limits at Infinity

Goal: Understand the end behavior (asymptotic behavior) of functions.

Definition. We say:

$$\lim_{x\to\infty} f(x) = L$$

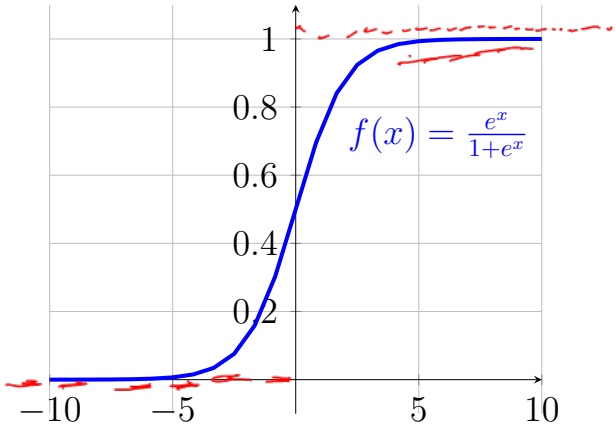
to mean $f(x)$ goes to L as x "goes to infinity" (i.e., x gets very, very large). Similarly,

$$\lim_{x\to-\infty} f(x) = L$$

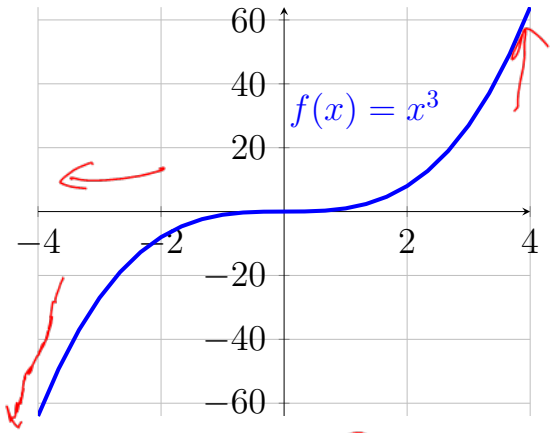
means $f(x)$ goes to L as x "goes to negative infinity" (i.e., x gets very, very "largely negative").

NOTE: A number being "small" typically refers to when it is close to 0. Hence, why I am using the phrase "largely negative," even though it is a bit awkward.

Example 1. For each part, determine $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$, where the graph of f is given.



$\lim_{x\to\infty} f(x) =$ 1



$\lim_{x\to\infty} f(x) =$ ∞

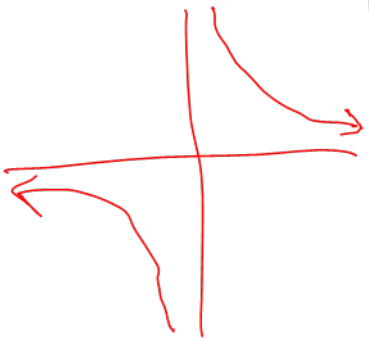
$\lim_{x\to-\infty} f(x) =$ 0

$\lim_{x\to-\infty} f(x) =$ $-\infty$

$\lim_{x\to\infty} f(x)$

Example 2. Determine the end behavior (limits at infinity) for the function $f(x) = \frac{1}{x}$

$f(x) = \frac{1}{x}$



x	$f(x)$
-1	-1
-100	$\frac{1}{100} = 0.01$
-10000	$\frac{1}{10000} = -0.0001$
-1000000	-0.000001

$\lim_{x\to\infty} f(x) = 0$

$\lim_{x\to-\infty} f(x) = 0$

In general, for any positive integer n,

$\frac{1}{x^2}, \frac{1}{x^5}$

$$\lim_{x\to\infty} \frac{1}{x^n} = \lim_{x\to-\infty} \frac{1}{x^n} = 0$$

Example 3. Compute $\lim_{x\to\infty} \left(2 + \frac{3}{x^2}\right)$ $\frac{1}{x^2}$

$$\lim_{x\to\infty} \left(2 + \frac{3}{x^2}\right) = \lim_{x\to\infty} 2 + \lim_{x\to\infty} \frac{3}{x^2}$$

$$= \underbrace{\lim_{x\to\infty} 2}_{2} + 3 \cdot \underbrace{\lim_{x\to\infty} \frac{1}{x^2}}_{0}$$

$$= 2$$

Rational Functions

Main Strategy (Formally): Factor out the highest power of x in the denominator

$\frac{x^2-1}{2x^2+1}$

Example 4. For $f(x) = \dfrac{x^2-1}{2x^2+1}$, compute $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$.

$$\lim_{x\to\infty} \frac{x^2-1}{2x^2+1} = \lim_{x\to\infty} \frac{x^2\left(\frac{x^2}{x^2} - \frac{1}{x^2}\right)}{x^2\left(\frac{2x^2}{x^2} + \frac{1}{x^2}\right)} = \lim_{x\to\infty} \frac{1 - \frac{1}{x^2}}{2 + \frac{1}{x^2}}$$

($\frac{1}{x^2} \to 0$)

$$= \frac{1-0}{2+0} = \frac{1}{2}$$

$$\lim_{x\to-\infty} \frac{x^2-1}{2x^2+1} = \lim_{x\to-\infty} \frac{1-\frac{1}{x^2}}{2+\frac{1}{x^2}} = \frac{1}{2}$$

$\frac{x^2+2}{x^3+x^2-1}$

Example 5. For $f(x) = \dfrac{x^2+2}{x^3+x^2-1}$, compute $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$.

$$\lim_{x\to\infty} \frac{x^2+2}{x^3+x^2-1} = \lim_{x\to\infty} \frac{x^3\left(\frac{x^2}{x^3} + \frac{2}{x^3}\right)}{x^3\left(\frac{x^3}{x^3} + \frac{x^2}{x^3} - \frac{1}{x^3}\right)}$$

$$= \lim_{x\to\infty} \frac{\frac{1}{x} + \frac{2}{x^3}}{1 + \frac{1}{x} - \frac{1}{x^3}} = \frac{0}{1} = 0$$

($\frac{1}{x} \to 0$, $\frac{2}{x^3} \to 0$; $1 \to 1$, $\frac{1}{x} \to 0$, $\frac{1}{x^3} \to 0$)

$$\lim_{x\to-\infty} \frac{x^2+2}{x^3+x^2-1} = \lim_{x\to-\infty} \frac{\frac{1}{x} + \frac{2}{x^3}}{1 + \frac{1}{x} - \frac{1}{x^3}} = \frac{0}{1} = 0$$

$\frac{x^2+x}{-x+3}$

Compare the ratios of the leading terms

Main Strategy (Informally):

Example 6. For $f(x) = \dfrac{x^2+x}{-x+3}$, compute $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$.

$\lim_{x\to\infty} \frac{x^2+x}{-x+3} = \lim_{x\to\infty} \frac{x^2}{-x} = \lim_{x\to\infty} -x = -\infty$

$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{x^2}{-x} = \lim_{x\to-\infty} -x = \infty$

In Summary: For a rational function:

$$f(x) = \frac{a(x)}{b(x)}$$

if $LT(a(x))$ and $LC(a(x))$ denote the leading term and leading coefficient of the polynomial $a(x)$ respectively, then,

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{LT(a(x))}{LT(b(x))}$$

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{LT(a(x))}{LT(b(x))}$$

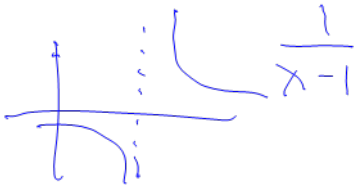
$3x^2+1 = a(x)$

$LT(a(x)) = 3x^2$

$LC(a(x)) = 3$

In particular,

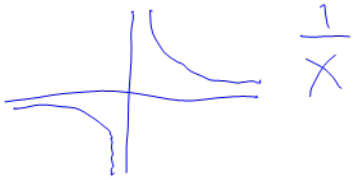
1. If $\deg(a(x)) = \deg(b(x))$, then:
$$\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = \frac{LC(a(x))}{LC(b(x))}$$
2. If $\deg(a(x)) < \deg(b(x))$, then:
$$\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = 0$$
3. If $\deg(a(x)) > \deg(b(x))$, more work is needed (since it depends on whether $\deg(a(x)) - \deg(b(x))$ is even or odd as well as whether $\dfrac{LC(a(x))}{LC(b(x))}$ is positive or negative).



Horizontal and Vertical Asymptotes (HAs and VAs)

Definition. A function $f(x)$ has a **vertical asymptote at** $x = c$ if either $\lim_{x\to c^-} f(x)$ or $\lim_{x\to c^+} f(x)$ is NOT finite.

Definition. A function $f(x)$ has a **horizontal asymptote at** $y = L$ if either $\lim_{x\to-\infty} f(x) = L$ or $\lim_{x\to\infty} f(x) = L$.



Example 7. Find the locations of any Horizontal and Vertical Asymptotes for $f(x) = \dfrac{3x^2+5}{x^2-4}$

$3x^2+5 \,/\, (x^2-4)$

$\frac{x^2}{x^3}$

For a rational function, VAs occur when the denominator is 0, but the numerator is non-zero. [After removing common factors]

$x^2 - 4 \overset{\text{set}}{=} 0$

$x^2 = 4$

$\Rightarrow x = -2, 2$

Numerator at -2 and 2 not zero $\Longrightarrow$ VAs at $x = -2, 2$

For a rational function $f(x)$, HAs occur when either $\lim_{x\to-\infty} f(x)$ or $\lim_{x\to\infty} f(x)$ is finite.

NOTE: Rational functions have at most one HA.

$$\lim_{x\to\infty} \frac{3x^2+5}{x^2-4} = \lim_{x\to\infty} \frac{3x^2}{x^2} = \lim_{x\to\infty} 3 = 3$$

I.e. $\lim_{x\to-\infty} f(x) = 3$. HA at $y = 3$

Slant (Oblique) Asymptotes

Recall Polynomial Long Division:

Theorem (Polynomial Division Algorithm)**.** For a rational function $\frac{a(x)}{b(x)}$ there are unique polynomials $q(x)$ (called the **quotient**) and $r(x)$ (called the **remainder**) such that:

$$\frac{a(x)}{b(x)} = q(x) + \frac{r(x)}{b(x)} \quad \text{AND} \quad \deg(r(x)) < \deg(b(x))$$

Compare this with numbers

$$\begin{array}{r} 3 \\ 2\overline{)7} \\ -6 \\ \hline 1 \end{array}$$

$$7 = 3(2) + 1$$

$$\frac{7}{2} = 3 + \frac{1}{2}$$

Example 8. Find the quotient and remainder when $2x^2 + 3x + 1$ is divided by $x - 1$.

$$\begin{array}{r} 2x + 5 \\ x-1\overline{)2x^2 + 3x + 1} \\ -(2x^2 - 2x) \\ \hline 5x + 1 \\ -(5x - 5) \\ \hline 6 \end{array}$$

$\deg(6) = 0 < 1 = \deg(x-1)$

$$\frac{2x^2+3x+1}{x-1} = (2x+5) + \frac{6}{x-1}$$

Definition. The line $y = mx + b$ is a **slant (oblique) asymptote** of the graph of $f(x)$ if:

$$\lim_{x\to\infty}[f(x) - (mx+b)] = 0 \quad \text{OR} \quad \lim_{x\to-\infty}[f(x) - (mx+b)] = 0$$

Informally, this means the graph of $f(x)$ gets closer and closer to the line $y = mx + b$ as x goes to positive or negative infinity.

Theorem. The following are equivalent for a rational function $\frac{a(x)}{b(x)}$:

1. $\frac{a(x)}{b(x)}$ has a slant asymptote.
2. The quotient in polynomial long division is a degree 1 polynomial.
3. $\deg(a(x)) = \deg(b(x)) + 1$

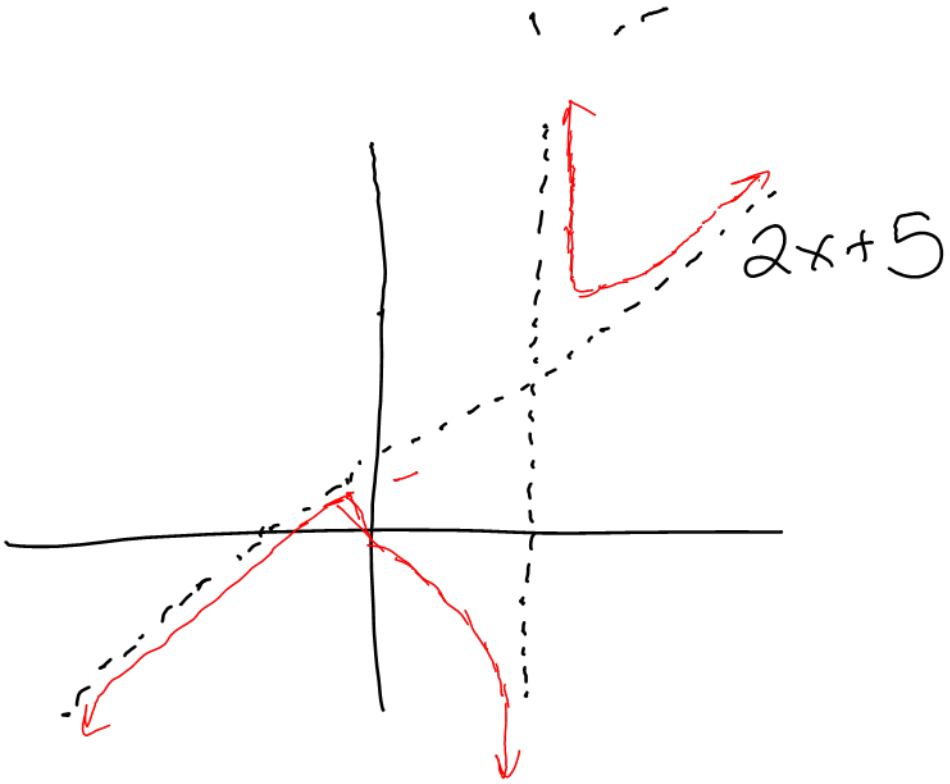
If these hold, the quotient obtained from long division is a slant asymptote.

Example 9. Does the function $f(x) = \frac{2x^2+3x+1}{x-1}$ have a slant asymptote? If so, find it.

Yes, since deg(bottom) = 1 and deg(top) = 2

$2 - 1 = 1$

$$2x^2 + 3x + 1 = \underbrace{[2x+5]}_{\text{Slant Asymptote}} + \underbrace{\frac{6}{x-1}}_{\to 0 \text{ as } x \to \pm\infty}$$



$\frac{2x^3-3x+1}{x^2-x-6}$

Example 10. Find all asymptotes (HAs, VAs, and slant) for the function $f(x) = \dfrac{2x^3 - 3x + 1}{x^2 - x - 6}$

HAs: None b/c we have a slant asy.

Slant:

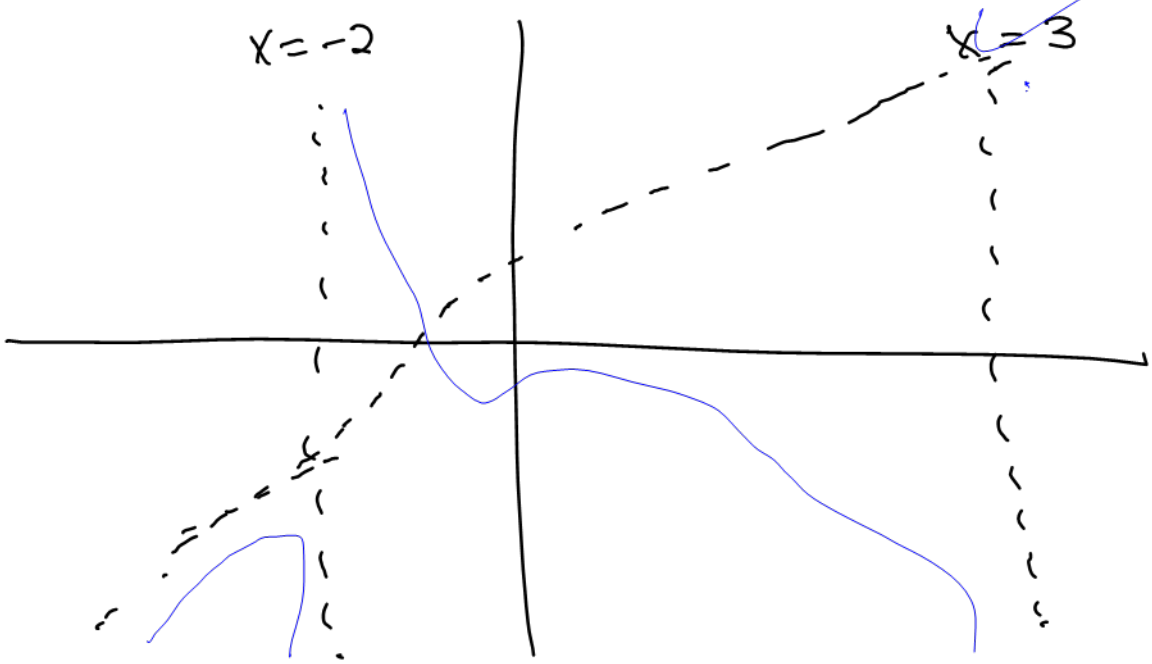
$$\begin{array}{r} 2x+2 \\ x^2-x-6 \overline{)\,2x^3-3x+1} \\ -(2x^3-2x^2-12x) \\ \hline 2x^2+9x+1 \\ -(2x^2-2x-12) \\ \hline 11x+13 \end{array}$$

← Slant Asymptote

EDITOR'S NOTE: Was originally -2x^2. Should be just 2x^2 →

VAs: Solve $x^2 - x - 6 = 0$

$\Rightarrow x = -2, 3$


x = -2
x = 3